Communication

Date:	21 October 2026	**Team ID:**	Piyush-Balodiya-GenAI-2026
Project Name:	Multimodal Construction Intelligence Platform (CONSTRUCTAI)	**Maximum Marks:**	1 Mark

Communication Plan:

Effective communication is essential for a successful project demonstration. This document outlines the communication strategy used within the team and with stakeholders throughout the project lifecycle, including how updates, issues, and feedback were managed.

S.No	Communication Type	Frequency	Channel / Tool	Participants	Purpose & Deliverable
1	**Team Standup**	Daily	Slack / Discord & Notion Workboard	Developer (Piyush Balodiya)	Review daily sprint backlog, track Groq API quota, and sync completed vision inference tasks.
2	**Progress Update**	Weekly	GitHub Projects & Email Summary	Developer & Project Evaluators	Share milestone progress reports, commit changelogs, and demo video screen recordings.
3	**Issue / Bug Discussion**	As Needed	GitHub Issues & Debug Console	Developer & Technical Mentors	Troubleshoot FastAPI payload bottlenecks, Groq vision prompt edge cases, and CORS bugs.
4	**Stakeholder Review**	Bi-Weekly	Google Meet & Live Screen Share	Developer, Mentors & NASSCOM Guides	Validate IS 456 concrete crack logic compliance and gather feedback on outdoor UI design.
5	**Final Demo Rehearsal**	Once	Local Staging Environment	Developer & Peer Evaluators	Full end-to-end dry run: live image upload, sub-second Groq inference, and automated PDF export.
6	**Industry Guide Feedback**	Bi-Weekly	Technical Sync Calls & WhatsApp	Civil Engineers & EHS Safety Experts	Calibrate real-world OSHA compliance rules, helmet color recognition, and crack severity thresholds.

Communication Challenges & Resolutions:

S. No	Challenge Faced	Resolution / Action Taken
1	**Intermittent Site Connectivity & Offline Sync** **Real-time communication dropouts on active construction site slabs.**	Engineered an offline-first caching mechanism in SQLite and localStorage for cached IS 456 remediation guidelines, with automatic asynchronous sync once internet is restored.
2	**Cross-Domain Terminology Mismatch** **AI vision technical jargon vs Civil Engineering structural codes.**	Standardized a shared dictionary mapping LLaMA 3.2 Vision tags directly into standardized Indian Standard (IS 456:2000) crack types and OSHA PPE violation codes.
3	**Asynchronous Query & API Latency Transparency** **Stakeholders unaware of inference status during heavy multi-MB image parsing.**	Implemented interactive real-time loading skeletons, step-by-step progress toast indicators, and sub-second Groq LPU throughput metrics on the React frontend.